CHAPTER 05

# Earth's Surface

Mountains, canyons, earthquakes, volcanoes — Earth's surface is always changing, just slowly enough that we rarely notice. In this chapter you'll dig from the crust down to the core, watch rocks transform, and discover the giant moving plates that reshape our planet.

📚 Georgia Standard S6E5

## WHAT YOU'LL LEARN

- **01** Inside the Earth
- **02** Minerals & Rocks
- **03** The Rock Cycle
- **04** Weathering, Erosion & Deposition
- **05** Plate Tectonics
- **06** Fossils & Soil

## 5.1 Inside the Earth

If you could slice Earth open like an apple, you'd see it's built in **layers**. We live on the thin outer skin, but most of our planet is hidden deep below — and it gets hotter and denser the deeper you go.

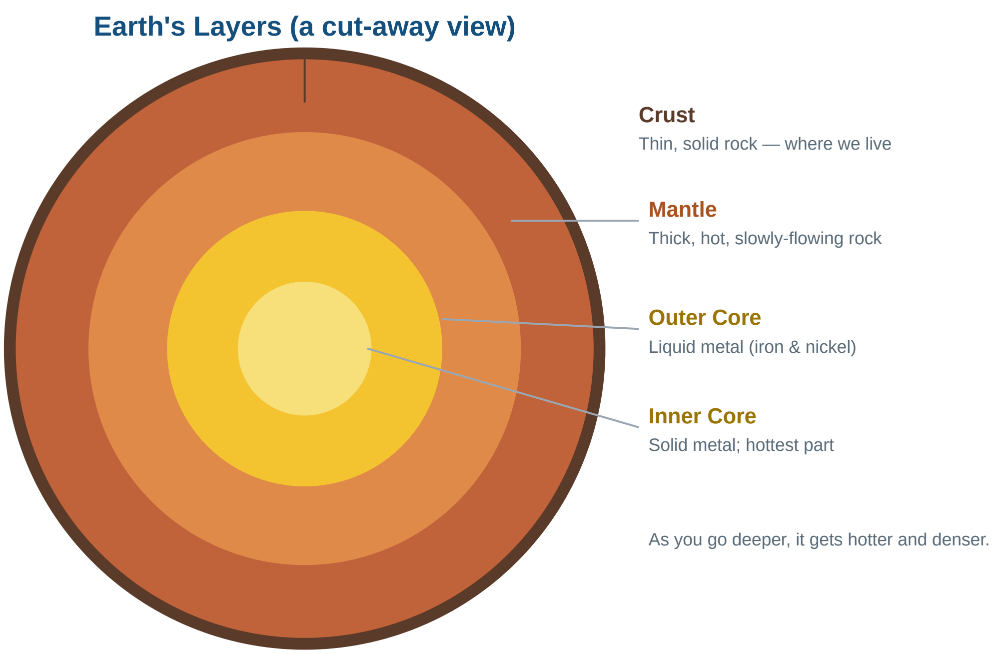



*Figure 5.1 — Earth's four main layers. The crust is thin like an apple's skin; the core is as hot as the surface of the Sun.*

| Layer | What It's Like | State |
|---|---|---|
| **Crust** | Thin, rocky outer shell where we live; thinnest layer | Solid |
| **Mantle** | Thick layer of hot rock that flows very slowly; the thickest layer | Solid but slowly flowing |
| **Outer Core** | Made of iron and nickel metal, hot enough to be liquid | Liquid |
| **Inner Core** | Iron and nickel; squeezed solid by huge pressure; hottest part | Solid |

### ★ Three Things That Change With Depth

As you travel from the crust toward the core, three things go **up**: **temperature** (hotter), **pressure** (more squeezing), and **density** (heavier, more tightly packed material). The inner core is nearly as hot as the surface of the Sun.

### ? Quick Check 5.1

1. Which layer do we live on? Which layer is the thickest?
2. Why is the inner core solid even though it's the hottest layer?

## 5.2 Minerals & Rocks

Rocks are all around you, but what are they actually made of? The answer is **minerals**. Understanding minerals is the first step to understanding rocks.

### What Is a Mineral?

A **mineral** is a naturally formed solid with a specific chemical makeup and an orderly structure. Quartz, gold, and salt are all minerals. Geologists identify minerals by testing properties like:

| Property | What It Tells You |
|---|---|
| **Hardness** | How easily it scratches (diamond is the hardest mineral) |
| **Color & Streak** | Its color, and the color of its powder when scraped |
| **Luster** | How shiny it is (metallic, glassy, dull) |
| **Cleavage** | How it breaks (smooth flat faces, or rough edges) |

### From Minerals to Rocks

A **rock** is a solid material made of one or more minerals mixed together. If minerals are the ingredients, a rock is the finished recipe. Granite, for example, is a rock made of several minerals — including quartz and feldspar — all locked together.

> **Mineral vs. Rock:** a mineral is a single pure ingredient with one recipe; a rock is a mixture of minerals. All rocks are made of minerals, but a mineral is not a rock.

**? Quick Check 5.2**

1. What is the difference between a mineral and a rock?
2. Name two properties geologists use to identify minerals.

## 5.3 The Rock Cycle

Rocks may seem permanent, but over millions of years any rock can slowly change into a different kind of rock. This never-ending process is called the **rock cycle**. There are three main types of rock, sorted by **how they form**.

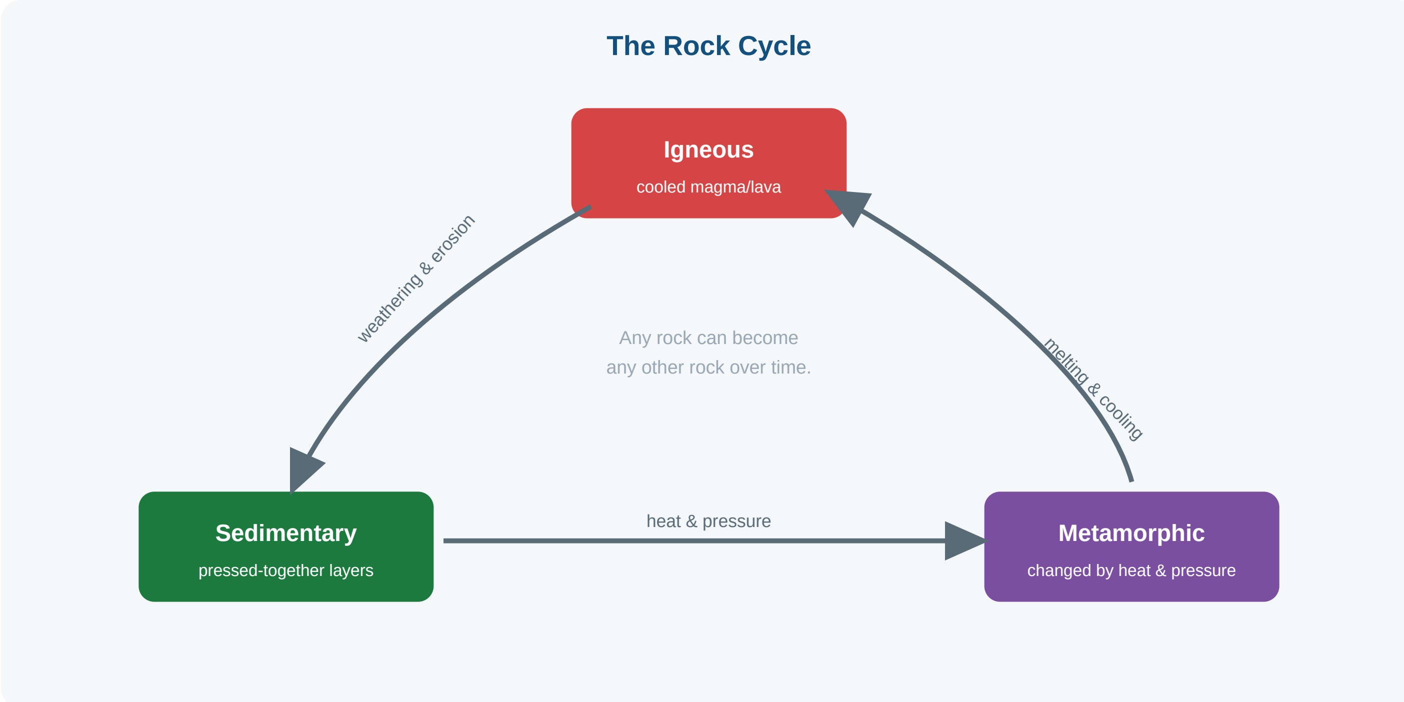



*Figure 5.2 — The rock cycle. Heat, pressure, melting, cooling, weathering, and erosion can turn any rock type into another.*

| Rock Type | How It Forms | Example |
|---|---|---|
| **Igneous** | Melted rock (magma or lava) cools and hardens | Granite, obsidian, basalt |
| **Sedimentary** | Bits of rock and shell settle in layers and get pressed together | Sandstone, limestone |
| **Metamorphic** | An existing rock is changed by intense heat and pressure (without melting) | Marble (from limestone), slate |

### ★ Decode the Names

The names give hints! **Igneous** comes from a word for "fire" (it forms from heat). **Sedimentary** comes from "sediment" (settling bits). **Metamorphic** means "changed form" (like "metamorphosis," when a caterpillar changes into a butterfly).

### ? Quick Check 5.3

1. What are the three main types of rock?
2. How does an igneous rock form?
3. Which rock type forms from layers of sediment pressed together?

## 5.4 Weathering, Erosion & Deposition

Three connected processes constantly reshape the land. They happen in order, like a three-step assembly line that takes rock apart in one place and rebuilds it somewhere else.


*Figure 5.3 — Weathering breaks rock down, erosion carries the pieces away, and deposition drops them in a new place.*

| Step | What Happens | Examples |
|---|---|---|
| **Weathering** | Rock is broken into smaller pieces, right where it sits | Ice cracking rock; roots splitting stone; acid rain dissolving rock |
| **Erosion** | The broken pieces are carried away by an agent of transport | Rivers, wind, glaciers, and waves moving sediment |
| **Deposition** | The pieces are dropped and settle in a new location | Sand building a beach; silt forming a river delta |

**Agents of Erosion**

The main "movers" that cause erosion are **water** (rivers, rain, waves), **wind**, **ice** (glaciers), and **gravity** (landslides). Moving water is the most powerful of all — it carved the Grand Canyon over millions of years.

**Don't Confuse These**

**Weathering** breaks rock apart but does *not* move it. **Erosion** is the *moving* part. An easy way to remember: **w**eathering = **w**ear down in place; **e**rosion = **e**xit, it moves away.

## Natural Processes and Human Activity

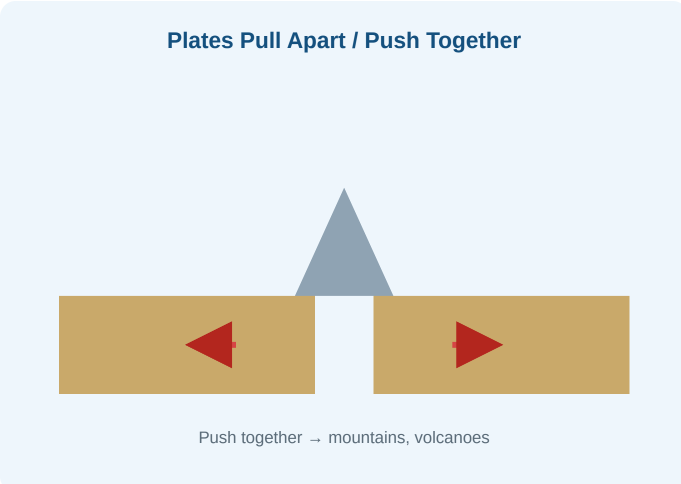
Both nature and people change Earth's surface. Natural forces like rivers, wind, and glaciers reshape land slowly over time. But **humans** can change the surface quickly — by mining, building cities and roads, farming, and clearing forests. Some human changes (like terracing hillsides) reduce erosion; others (like removing plants) speed it up.

**Quick Check 5.4**

1. Put these in order: deposition, weathering, erosion.
2. What is the difference between weathering and erosion?
3. Name one way humans change Earth's surface.

## 5.5 Plate Tectonics

Earth's crust isn't one solid shell. It's cracked into giant pieces called **tectonic plates** that float on the slowly-flowing mantle below. These plates move just a few centimeters a year — about as fast as your fingernails grow — but over millions of years that movement builds mountains and triggers earthquakes and volcanoes.



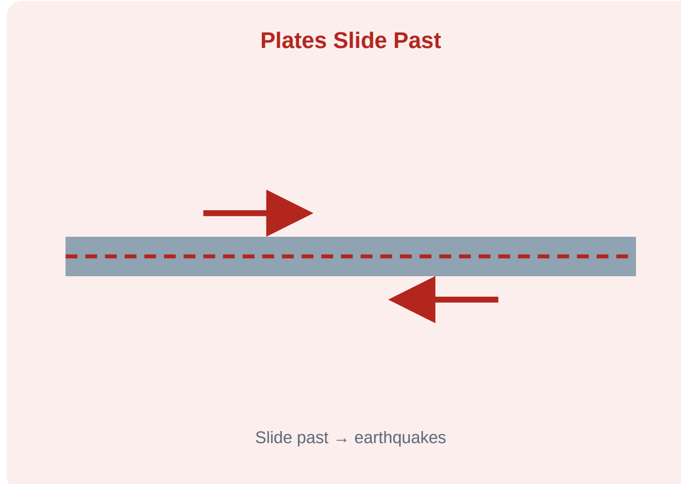




*Figure 5.4 — When plates push together, they build mountains and volcanoes. When they slide past each other, they cause earthquakes.*

| Plate Movement | What Happens | Result |
| --- | --- | --- |
| **Push together (collide)** | Crust crumples upward or one plate dives under another | Mountains and volcanoes |
| **Pull apart (separate)** | Magma rises to fill the gap and makes new crust | Mid-ocean ridges, rift valleys |
| **Slide past** | Plates grind and catch, then suddenly slip | Earthquakes |

### Earthquakes and Volcanoes

An **earthquake** is a sudden shaking of the ground when plates slip past each other and release built-up energy. A **volcano** is an opening where magma from the mantle reaches the surface (where it's called lava). Both happen most often at the edges of plates — which is why a "Ring of Fire" of earthquakes and volcanoes circles the Pacific Ocean, tracing the plate boundaries.

**? Quick Check 5.5**

1. What are tectonic plates, and what do they float on?
2. What happens when two plates push together?
3. What causes an earthquake?

## 5.6 Fossils & Soil

The ground beneath us keeps records of the past and supports the life of today. Two final pieces of Earth's surface tell important stories: **fossils** and **soil**.

## Fossils: Clues to Earth's Past

A **fossil** is the preserved remains or trace of a living thing from long ago, usually found in sedimentary rock. Fossils are powerful evidence: they show that Earth's surface and climate have **changed** over time. For example, fossils of **sea creatures** have been found on mountaintops — proof that land once underwater was pushed upward by moving plates. Tropical plant fossils in cold regions show the climate there was once warm.

### ★ What Fossils Tell Us

Because most fossils form in sedimentary layers, **deeper layers are usually older** and the fossils in them lived longer ago. By reading the layers like pages in a book, scientists piece together how life and the land have changed.

## Soil: Where Rock Meets Life

**Soil** is the loose top layer that forms when weathered rock mixes with decayed plant and animal material (called **organic matter** or humus). Soil forms in **layers**, called a soil profile.

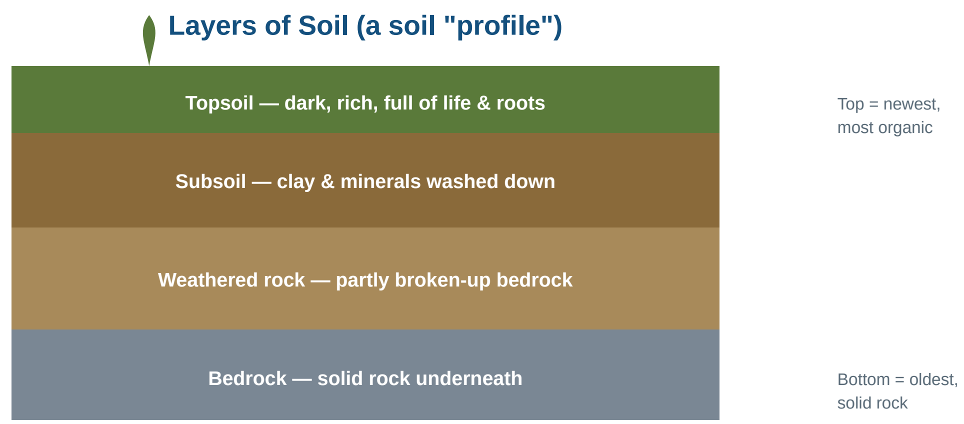



*Figure 5.5 — A soil profile. The dark, rich topsoil sits on top; solid bedrock lies at the bottom.*

The dark **topsoil** on top is rich in organic matter and is where most plants grow. Below it, the subsoil holds minerals washed down from above. Deeper still is weathered rock, and at the very bottom is solid **bedrock**. Healthy soil is a mix of weathered rock *and* decomposed living things — both are needed for plants to thrive.

### ? Quick Check 5.6

1. How do fossils of sea creatures on a mountaintop show that Earth's surface has changed?
2. What two main ingredients make up soil?
3. Which soil layer is richest for growing plants?

# Chapter 5 Wrap-Up

### ★ Key Takeaways

- **Earth has four layers:** crust, mantle, outer core, inner core. Deeper = hotter, denser, more pressure.
- **Minerals** are pure ingredients; **rocks** are mixtures of minerals. Three rock types: igneous, sedimentary, metamorphic.
- **The rock cycle** slowly turns any rock into another through heat, pressure, melting, weathering, and erosion.
- **Weathering** breaks rock down, **erosion** moves it, **deposition** drops it. Nature and humans both reshape the land.
- **Tectonic plates** move to build mountains and cause earthquakes/volcanoes. **Fossils** show the surface and climate have changed; **soil** forms in layers from weathered rock and organic matter.